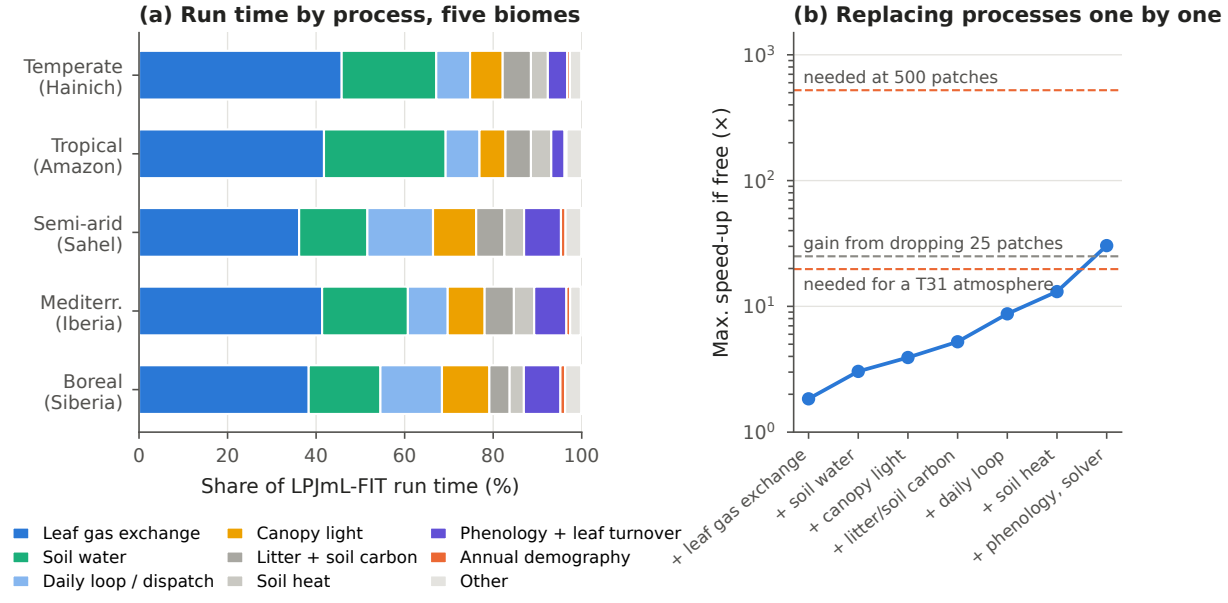
(a) Run time by process, five biomes
Temperate (Hainich)
Tropical (Amazon)
Semi-arid (Sahel)
Mediterr. (Iberia)
Boreal (Siberia)
0
20
40
60
80
100
Share of LPJmL-FIT run time (%)
(b) Replacing processes one by one
Max. speed-up if free (×)
10^3
10^2
10^1
10^0
needed at 500 patches
gain from dropping 25 patches
needed for a T31 atmosphere
+ leaf gas exchange
+ soil water
+ canopy light
+ litter/soil carbon
+ daily loop
+ soil heat
+ phenology, solver
Leaf gas exchange
Soil water
Daily loop / dispatch
Canopy light
Litter + soil carbon
Soil heat
Phenology + leaf turnover
Annual demography
Other